

Chemoprevention of breast cancer by dietary compounds.

Breast cancer is the leading cause of cancer-related deaths in women in the United States and many other countries. There is an immediate need for more effective and less toxic therapeutic and preventive strategies for many cancers, especially for breast cancer. Natural products are being tested with a hope of identifying novel potent molecules as anticancer agents. Phytochemicals and dietary compounds have been used for the treatment of various illnesses throughout history due to their safety, low toxicity, and general availability. Currently, many active phytochemicals are in clinical trials. Preclinical and clinical studies have indicated that daily consumption of dietary phytochemicals reduces the risk of several cancers. Phytochemicals can inhibit, delay, or reverse carcinogenesis by inducing detoxifying and antioxidant enzymes, by regulating inflammatory/proliferative signaling pathways, and by inducing apoptosis. This review article describes some of the potential natural cancer preventive compounds, along with a mechanistic discussion of their interactions with key cellular signal transduction pathways as well as their contribution to the suppression of breast cancer cell growth.